



PARAMETER LIST



AGC-4



1. General information

1.1 Warnings, legal information and safety	7
1.1.1 Warnings and notes	7
1.1.2 Legal information and disclaimer	7
1.1.3 Factory settings	7
1.2 About the Parameter list	8
1.2.1 Purpose of the Parameter list	8
1.2.2 Intended users	8

2. Alarm list

2.1 General information about the alarm list	9
2.1.1 Alarm list features and options	9
2.2 Protection parameters	11
2.2.1 Reverse power protections	11
2.2.2 Over-current protections	11
2.2.3 Voltage protections	14
2.2.4 Frequency protections	16
2.2.5 Busbar voltage protections	18
2.2.6 Busbar frequency protections	21
2.2.7 Mains failure protections	25
2.2.8 Overload protections	27
2.2.9 Current unbalance protection	28
2.2.10 Voltage unbalance protections	29
2.2.11 Reactive power import (loss of excitation) protection	29
2.2.12 Reactive power export (over-excitation) protection	30
2.2.13 Negative sequence	30
2.2.14 Zero sequence	31
2.2.15 Directional over-current protections	32
2.2.16 Busbar unbalance voltage	33
2.2.17 HVRT	33
2.2.18 Time-dependent under-voltage (LVRT)	33
2.2.19 Power-dependent reactive power import	35
2.2.20 Power-dependent reactive power export	35
2.2.21 Non-essential load trip (load shedding)	36
2.2.22 Under-voltage and reactive power low	38
2.3 Control parameters: Synchronisation	40
2.3.1 Synchronisation and breaker alarms	40
2.3.2 Mains sync. inhibit	43
2.4 Control parameters: Regulation	45
2.4.1 Regulation alarms	45
2.5 Input/output parameters: Binary input setup	47
2.5.1 Digital input 23-27 setup	47
2.5.2 Digital input 43-55 setup (requires option M12)	48
2.5.3 Digital input 91-97 setup (requires option M13.6)	53
2.5.4 Digital input 102-108 setup	55
2.5.5 Digital input 112-117 setup	56
2.5.6 Emergency stop	58
2.5.7 Digital input 127-133 setup (requires option M13.8)	58
2.5.8 M-Logic alarm 1-5 setup	61

2.6 Input/output parameters: Analogue input setup	63
2.6.1 Analogue input setup (requires option M15.6)	63
2.6.2 Analogue input setup (requires option M15.8)	67
2.6.3 Analogue input setup (requires option M16.6)	71
2.6.4 Analogue input setup (requires option M16.8)	80
2.7 Multi-functional analogue input setup	90
2.7.1 Multi-input no. 102	90
2.7.2 Multi-input no. 105	94
2.7.3 Multi-input no. 108	98
2.7.4 Speed and running feedback setup	103
2.7.5 Differential measurement	106
2.7.6 Aux. supply setup	113
2.8 System parameters: General setup	115
2.8.1 Stop coil wirebreak and internal communication alarms	115
2.8.2 Engine heater failure	115
2.8.3 Running detection	116
2.8.4 Battery tests	116
2.8.5 Max. ventilation	118
2.8.6 Switchboard error: Block and Stop	118
2.8.7 Switchboard error: Not in auto	119
2.8.8 Oil renewal	119
2.8.9 Avg U BB (with option A1)	120
2.9 System parameters: Communication	121
2.9.1 External communication error	121
2.9.2 Engine interface communication alarms	121
2.9.3 Power management communication error	126
2.9.4 Internal CAN communication error	126
2.10 External I/O parameters	128
2.10.1 External I/O alarm setup	128
2.10.2 Analogue inputs (requires option H8.x)	128
2.10.3 External analogue input scale (requires option H8.x)	129
2.10.4 Digital inputs (requires option H8.x)	129

3. System parameters

3.1 General setup	130
3.1.1 Nominal settings	130
3.1.2 Breaker control	132
3.1.3 Date and time	132
3.1.4 Master clock	133
3.1.5 Summer/winter time	133
3.1.6 Counters	133
3.1.7 Pulse counter	133
3.1.8 Timers	134
3.1.9 Command timers	136
3.1.10 Language	139
3.1.11 Alarm horn	140
3.1.12 Alarm jump	140
3.1.13 Diagnostics	140
3.1.14 I thermal demand	140

3.2 Genset setup	141
3.2.1 Genset mode	141
3.2.2 Test	141
3.2.3 Run coil setup	141
3.2.4 Running, start and stop	142
3.2.5 Idle start	143
3.2.6 Analogue load sharing lines output	144
3.2.7 Power derate	144
3.2.8 Cooling ventilation	146
3.2.9 Fan logic	146
3.2.10 Engine heater	148
3.2.11 Fuel transfer pump logic	149
3.2.12 Tank capacity	149
3.2.13 Digital AVR parameters	149
3.3 Engine interface communication	154
3.3.1 Engine interface communication	154
3.3.2 Cab Message 1 source address	155
3.3.3 Generator Control 1 source address	155
3.3.4 EIC derate	155
3.3.5 Caterpillar/Perkins	155
3.3.6 Isuzu	156
3.3.7 JCB	156
3.3.8 Kohler	156
3.4 Mains setup	157
3.4.1 Mains setup	157
3.4.2 Test	158
3.4.3 Controller settings	158
3.4.4 Mains failure	159
3.4.5 Y1(X1) droop curve	160
3.4.6 Y2(X2) droop curve	160
3.4.7 Power offset	161
3.4.8 Cos phi offset	161
3.4.9 Mains ATS function	161
3.4.10 Mains transducers	161
3.5 Power management setup	163
3.5.1 Power management general setup	163
3.5.2 Available power	166
3.5.3 Load-dependent start and stop	167
3.5.4 Priority selection	169
3.5.5 Internal CAN protocol	170
3.5.6 Power management internal communication	171
3.5.7 Ground relay	171
3.6 External communication setup	172
3.6.1 CAN port setup	172
3.6.2 Modbus/Profibus communication	172
3.6.3 External I/O communication setup	172
3.7 RMI inputs	174
3.7.1 RMI 102	174
3.7.2 RMI 105	175

3.7.3 RMI 108.....	175
3.7.4 Multi-input selections 102, 105, 108.....	177
3.7.5 Multi-input selections, option M16.6.....	177
3.7.6 Multi-input selections, option M16.8.....	178
3.7.7 4-20 mA input scaling.....	179
3.7.8 Parameter ID.....	179
3.8 External digital outputs.....	180
3.8.1 External digital outputs (requires option H8).....	180
3.8.2 External module status.....	180
3.8.3 Supervision.....	180
3.8.4 AC average.....	181
3.9 Jump menus.....	186
3.9.1 Software version.....	186
3.9.2 Display character test.....	186
3.9.3 Service port.....	186
3.9.4 Scaling.....	186
3.9.5 M4 software version.....	186
3.9.6 Device type.....	187
3.9.7 Password.....	187
3.9.8 Service menu.....	187
3.9.9 AC configuration.....	188
3.9.10 Angle compensation BB/G.....	188
3.9.11 Backlight dimmer.....	188
3.9.12 Application drawing.....	188
3.9.13 Internal CAN protocol.....	189
3.9.14 Quick setup.....	189
3.9.15 Application broadcast.....	190
3.9.16 Memory backup.....	190
3.9.17 Data logging.....	191
3.10 Utility software.....	192
3.10.1 GSM settings.....	192
4. Control parameters	
4.1 Synchronisation.....	193
4.2 Regulation.....	196
4.3 Output setup.....	204
4.3.1 Digital output setup.....	204
4.4 Analogue output.....	210
4.4.1 Regulation reference output.....	210
4.4.2 Analogue output limits.....	211
4.5 Transducer outputs.....	212
4.6 Analogue regulator output setup.....	218
4.6.1 Regulator output selection.....	218
5. Utility software settings	
5.1 Application supervision.....	219
5.1.1 Plant settings.....	219
5.2 Advanced protection.....	219
5.3 PID settings.....	220
5.4 CIO settings.....	221

5.5 RRCR settings..... 221

5.6 Counters..... 221

5.7 Identifiers..... 221

1. General information

1.1 Warnings, legal information and safety

1.1.1 Warnings and notes

Throughout this document, a number of warnings and notes with helpful user information will be presented. To ensure that these are noticed, they will be highlighted as follows in order to separate them from the general text.

Warnings



DANGER!

Warnings indicate a potentially dangerous situation, which could result in death, personal injury or damaged equipment, if certain guidelines are not followed.

Notes



INFO

Notes provide general information, which will be helpful for the reader to bear in mind.

1.1.2 Legal information and disclaimer

DEIF takes no responsibility for installation or operation of the generator set. If there is any doubt about how to install or operate the engine/generator controlled by the Multi-line 2 unit, the company responsible for the installation or the operation of the set must be contacted.



INFO

The Multi-line 2 unit is not to be opened by unauthorised personnel. If opened anyway, the warranty will be lost.

Disclaimer

DEIF A/S reserves the right to change any of the contents of this document without prior notice.

The English version of this document always contains the most recent and up-to-date information about the product. DEIF does not take responsibility for the accuracy of translations, and translations might not be updated at the same time as the English document. If there is a discrepancy, the English version prevails.

1.1.3 Factory settings

The Multi-line 2 unit is delivered from the factory with default settings. These are not necessarily correct for the engine/generator set. Check all the settings before running the engine/generator set.

1.2 About the Parameter list

1.2.1 Purpose of the Parameter list

The parameter list is a reference document. It includes the AGC-4 alarms and parameters. If your controller does not include the relevant option, the option-dependent parameters may not be accessible.

Parameters with numbers up to 9999 can generally be configured using the DU-2 display unit, TDU 107 or the utility software. Some of the JUMP menu parameters (9000 to 9250) can only be configured using the display unit.

Parameters with numbers above 10000 can only be configured from the TDU 107 or the utility software.

The utility software includes settings that do not have parameter numbers. These are described briefly under [Utility software settings](#).

For some of the parameters, the AGC-4 **Designer's reference handbook** and/or **Option** documents contain more information.

1.2.2 Intended users

This Parameter List is mainly intended for the person responsible for the unit parameter setup. In most cases, this would be a panel builder designer. Naturally, other users might also find useful information here.

2. Alarm list

2.1 General information about the alarm list

2.1.1 Alarm list features and options

In the following, these abbreviations are used:

- G: Generator
- GB: Generator breaker
- TB: Tie breaker (for mains unit)
- MB: Mains breaker
- BTB: Bus tie breaker
- BA: Busbar A (BTB unit)
- BB: Busbar (BTB unit: Busbar B)
- N/A: Not available

This chapter includes an alarm list, with the alarm options.

Each alarm can include the following parameters:

- Set point: The alarm set point. This is often a percentage of the nominal value.
- Delay: The timer setting is the time that must expire from the set point is reached until the alarm is activated.
- Relay output A: A relay can be activated by output A.
- Relay output B: A relay can be activated by output B.
- An alarm is activated if no relay output, A or B, is selected. Do not select Limits/Limit relay if you want an alarm to be raised together with a relay output A or B.
- Enable: The alarm can be activated or deactivated.
- Fail class: When the alarm occurs the unit will react depending on the selected fail class.

The fail classes are:

Fail class	DG (diesel generator)	Mains unit	BTB (bus tie breaker)
F1	Block	Block	Block
F2	Warning	Warning	Warning
F3	Trip GB	Trip TB	Trip BTB
F4	Trip + Stop	Trip MB	N/A
F5	Shutdown	N/A	N/A
F6	Trip MB	N/A	N/A
F7	Safety stop	N/A	N/A
F8	Trip MB/GB	N/A	N/A
F9	Controlled stop	N/A	N/A



INFO

Small differences due to the character of the parameters may exist between the individual alarms.

The utility software includes some alarm functions that are not available from the display unit. For example, it is possible to select automatic acknowledgement of the alarm.

Parameter "G -P> 1" (Channel 1000)

Setpoint :

-50 -5 % 0

Timer :

0,1 10 sec 100,0

Fail class : Trip of GB

Output A : Not used

Output B : Not used

Password level : Customer

☒ Enable
☐ High Alarm
☐ Inverse proportional

☐ Auto acknowledge
Inhibits...

Commissioning

Actual value : 0 %

Time elapsed : 0 sec (0 %)

0 sec 10 sec

Write OK Cancel